

Discussion

Synthesis of Findings

Our simulation-based evaluation across six multiagent architecture models validates the core theoretical claims of the Cognitive Integrity Framework established in Part 1. The 94% overall detection rate achieved by the full CIF deployment represents a substantial improvement over any individual defense mechanism, confirming that the multiplicative composition theorems translate from formal proofs to practical protection. More importantly, the consistency of this result across architecturally diverse systems—from hierarchical orchestrator patterns to peer-to-peer topologies—suggests that CIF's formal abstractions capture genuine structural properties of multiagent security rather than artifacts of specific implementation choices. We now examine these findings in detail, beginning with the mechanisms underlying layered defense success.

Why Layered Defense Succeeds

Defense Mechanism Detection Overlap

 Input-layer attacks  Belief-layer attacks  Identity-layer attacks  Behavioral attacks 

Limitations and Threats to Validity

Residual Attack-Type Vulnerabilities

Despite strong overall performance, specific attack types remain challenging and merit detailed examination.

Semantic equivalent attacks pose the most significant residual risk.

When an adversary rephrases a known injection to preserve its semantic intent while altering surface-level features, pattern-matching defenses fail to recognize the attack. Our evaluation shows that the CIF firewall's TF-IDF and embedding-based classifiers achieve 89% detection on direct injections but only 72% on semantically equivalent reformulations. This gap is not unique to CIF; it reflects a fundamental limitation of feature-based detection that also affects commercial tools such as Lakera Guard and LLM Guard. Incorporating large language model-based semantic analysis into the firewall classification pipeline represents the most promising mitigation, though it introduces additional latency and cost trade-offs.

Progressive drift attacks exploit the tension between detection sensitivity and observation window length. An attacker who

Relationship to Prior Work

Our empirical results contextualize CIF's contributions relative to the related work surveyed in sec:related-work. Three findings merit specific comparison.

First, CIF's cognitive firewall achieves 85% detection on prompt injection when deployed alone—comparable to published detection rates for commercial single-agent tools such as NeMo Guardrails [?] and Lakera Guard—but the full CIF stack reaches 96% by composing the firewall with mechanisms targeting attack vectors that single-agent tools do not address (trust exploitation, belief manipulation, coordination). This validates the central thesis that multiagent security requires defenses beyond input filtering. A head-to-head comparison on standardized benchmarks (e.g., StrongReject) remains an important direction for future work.

Second, our trust calculus with δ^d decay is, to our knowledge, the first formally verified bound on delegated trust in LLM-based agent systems. While classical trust frameworks (FIRE [?], REGRET [?]) address trust propagation in distributed systems, none provide the

Open Research Directions

Adversarial Retraining and Honeypot Agents

Detection rates inevitably degrade as adversaries learn to evade deployed defenses—a dynamic well-characterized by the arms race model in security research and formalized in Part 1's detection degradation analysis (Section 4). Our current evaluation uses a static attack corpus, which represents a snapshot of adversarial capability rather than an evolving threat. Future work should investigate adversarial retraining of CIF detection mechanisms, where the firewall and anomaly detectors are periodically retrained on attacks that successfully evade current configurations. This approach, analogous to adversarial training in machine learning robustness research, requires careful management of the retraining loop to prevent catastrophic forgetting of previously effective detection patterns.

A second promising direction is the deployment of honeypot agents—intentionally vulnerable agents designed to attract and characterize novel attack techniques without risking production systems. By analyzing the attacks directed at honeypot agents,